Lecture Notes: General Relativity

Part I: Chapter 2 - From Topological to Differentiable Manifolds

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Chapter 1

From Topological to Differentiable Manifolds

While general topological spaces can be extremely wild and unpredictable, physicists typically study a very well-behaved class of spaces called **Topological Manifolds**. A topological space (M, T) is an n-dimensional manifold if it satisfies three strict mathematical axioms, each holding profound implications for physical reality.

1.1 The Three Axioms of a Manifold

1. **Hausdorff Space:** Any two distinct points can be strictly separated by disjoint open sets.

$$\forall p, q \in M, p \neq q, \exists U_p, U_q \in T \text{ s.t. } p \in U_p, q \in U_q, \text{ and } U_p \cap U_q = \emptyset$$

2. **Locally Euclidean:** Every point has a local neighborhood that looks exactly like flat n-dimensional space.

$$\forall p \in M, \exists U_p \in T \text{ containing } p, \text{ and a homeomorphism } \phi : U_p \to V \subset \mathbb{R}^n$$

   (where V is an open set in the usual topology of $\mathbb{R}^n$).

3. **Second-Countable:** The topology must possess a countable basis $\mathcal{B}$. This ensures the space isn't "too massive" (which successfully rules out pathological spaces like the discrete topology on an uncountable set, which would yield $2^{|X|}$ open sets).

1.2 Physical Meaning: Determinism and Observers

Why the Hausdorff Axiom? Consider a hypothetical space that fails the Hausdorff condition, such as the **"Line with Two Origins"** (defined as $(\mathbb{R} \setminus \{0\}) \cup \{0_A, 0_B\}$). In this pathological space, the two origins 0_A and 0_B cannot be separated by disjoint open sets. Physically, if a particle's trajectory approaches zero, it would non-deterministically "branch" into either 0_A or 0_B. Because classical physics (including GR) is fundamentally deterministic, the Hausdorff condition guarantees that limits of trajectories are unique, thereby preventing physical paradoxes and the breakdown of causality.

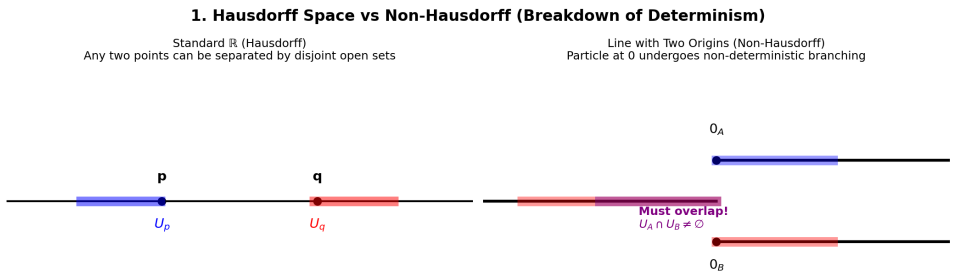
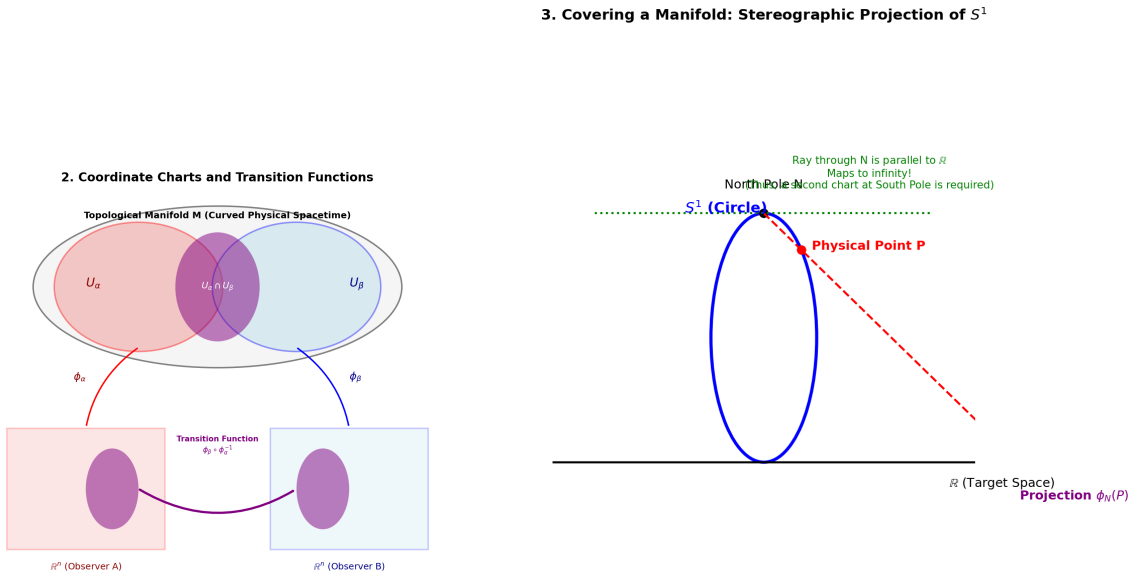


Figure 1.1: Hausdorff Space: Two distinct points can be separated by disjoint open neighborhoods, preserving determinism.

paragraphWhy Locally Euclidean? Due to the finite speed of light, no physical observer can have a "global" instantaneous view of the entire universe. A physical observer can only measure a restricted, local patch of spacetime (U_α). Therefore, the homeomorphism $\phi_\alpha : U_\alpha \to \mathbb{R}^n$ represents the observer setting up a local **coordinate system** to take physical measurements.

1.3 Coordinate Charts and Transition Functions

If observer A sees patch U_α and assigns coordinates via ϕ_α, and observer B sees an overlapping patch U_β and assigns coordinates via ϕ_β, they will inevitably record different numerical values for the exact same physical event $P \in U_\alpha \cap U_\beta$.



(a) Coordinate Charts

(b) Stereographic Projection

To ensure the laws of physics are absolutely consistent between them, we need a smooth mathematical method to translate between their differing coordinates. This operation is governed by the **Transition Function** (or coordinate transformation):

$$\phi_\beta \circ \phi_\alpha^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \to \phi_\beta(U_\alpha \cap U_\beta)$$

Because both ϕ_α and ϕ_β are homeomorphisms mapping to $\mathbb{R}^n$, the transition function is uniquely a mapping strictly from $\mathbb{R}^n \to \mathbb{R}^n$. Consequently, this allows physicists to use standard multivariable calculus and tensor analysis to relate different observers' measurements across curved spacetime.

1.4 Example: The n-Sphere (S^n)

Consider the n-dimensional sphere embedded in $\mathbb{R}^{n+1}$:

$$S^n = \{x \in \mathbb{R}^{n+1} \mid x^2 = 1\}$$

S^n is a topological manifold. It naturally inherits its topology directly from $\mathbb{R}^{n+1}$ via the **Subspace Topology** ($T_{S^n} = \{U \cap S^n \mid U \in T_{\mathbb{R}^{n+1}}\}$).

However, you **cannot** cover the entire sphere with just one single coordinate chart (because S^n is topologically not homeomorphic to $\mathbb{R}^n$). For S^1 (a circle) or S^2 (a spherical surface), we must use at least two charts to map the manifold completely. A standard method is **Stereographic Projection**:

- Chart 1: Remove the North Pole N. $\phi_N : S^n \setminus \{N\} \to \mathbb{R}^n$
- Chart 2: Remove the South Pole S. $\phi_S : S^n \setminus \{S\} \to \mathbb{R}^n$

Together, these two overlapping open sets cover the entire sphere without hitting the infinite singularity at the poles, successfully satisfying the locally Euclidean requirement for the manifold.

1.5 The Limits of Topological Manifolds: Orbifolds

Before we introduce calculus on manifolds, we must recognize that not all spaces constructed by identifying points result in a smooth, well-behaved manifold. A classic example encountered frequently in physics is the **Orbifold**, specifically formed by taking a flat space and applying a discrete symmetry group.

Consider the quotient space $\mathbb{R}^n/\mathbb{Z}_2$. Here, we identify every physical point x with its reflection $-x$.

- **For $n = 2$ ($\mathbb{R}^2/\mathbb{Z}_2$):** Visually, identifying opposite points on a plane folds it into a **cone**. Topologically, you can "flatten" this cone, meaning it is technically *homeomorphic* to $\mathbb{R}^2$. However, the apex of the cone forms a sharp geometric singularity. Because of this sharp tip, we cannot uniquely define a tangent plane, meaning it is *not* smoothly equivalent to $\mathbb{R}^2$.
- **For $n = 3$ ($\mathbb{R}^3/\mathbb{Z}_2$):** If we carve out a small neighborhood around the singular origin, $B((0,0,0),\epsilon)$, the remaining space is topologically equivalent to the Real Projective Plane ($\mathbb{RP}^2$). At the origin itself, the local neighborhood does not look like $\mathbb{R}^3$ at all, meaning it fails the "locally Euclidean" axiom entirely.

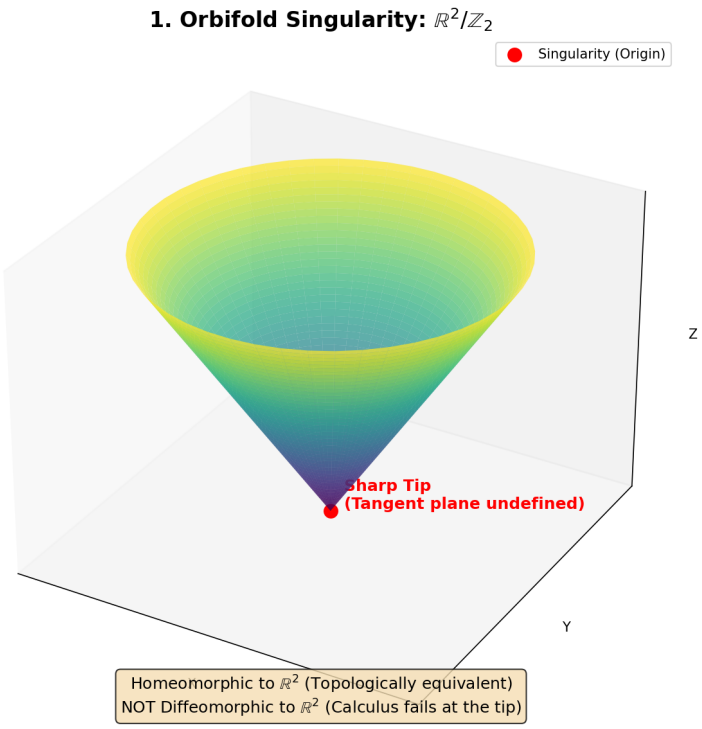


Figure 1.2: Orbifold Singularity: $\mathbb{R}^2/\mathbb{Z}_2$ forms a cone with a sharp tip at the origin.

1.6 Upgrading to Differentiable Manifolds

In physics, merely knowing the topology of spacetime is insufficient. To calculate physical observables or write down equations of motion, we need to perform calculus. A pure topological manifold requires its transition functions to be merely *continuous* (C^0), which does not support calculus. We must upgrade it to a **Differentiable Manifold**.

We impose a strict new condition: the transition functions between overlapping charts must be **smooth** (infinitely differentiable).

$$\phi_\beta \circ \phi_\alpha^{-1} \in C^\infty$$

textbfTerminology Check:

- A single pair (U_α, ϕ_α) is called a **Coordinate Chart** (mathematicians) or a **Coordinate System** (physicists).
- A complete collection of these charts that entirely covers the manifold M is called an **Atlas** $\{(U_\alpha, \phi_\alpha)\}$.

1.6.1 Compatible Charts and the Maximal Atlas

Suppose two physicists define a smooth structure on the same manifold M. Physicist A chooses an open cover with charts $\{U_\alpha, \phi_\alpha\}$, and Physicist B chooses a different cover $\{U'_\beta, \phi'_\beta\}$. If the cross-transition functions between their charts are smooth (C^∞), the two atlases are **compatible**.

To avoid ambiguity, a differentiable manifold is defined using a **Maximal Atlas**, which includes *all possible* compatible coordinate charts. Physically, this means including every conceivable valid observer whose coordinate transformations smoothly agree with the rest.

subsectionThe Anomaly of Exotic $\mathbb{R}^4$ Does a unique topological manifold always lead to a unique differentiable manifold? No. If a topological manifold M is homeomorphic to $\mathbb{R}^n$, is it also diffeomorphic to $\mathbb{R}^n$?

- For $n \neq 4$: Yes. Any smooth manifold homeomorphic to $\mathbb{R}^n$ is inherently diffeomorphic to it.

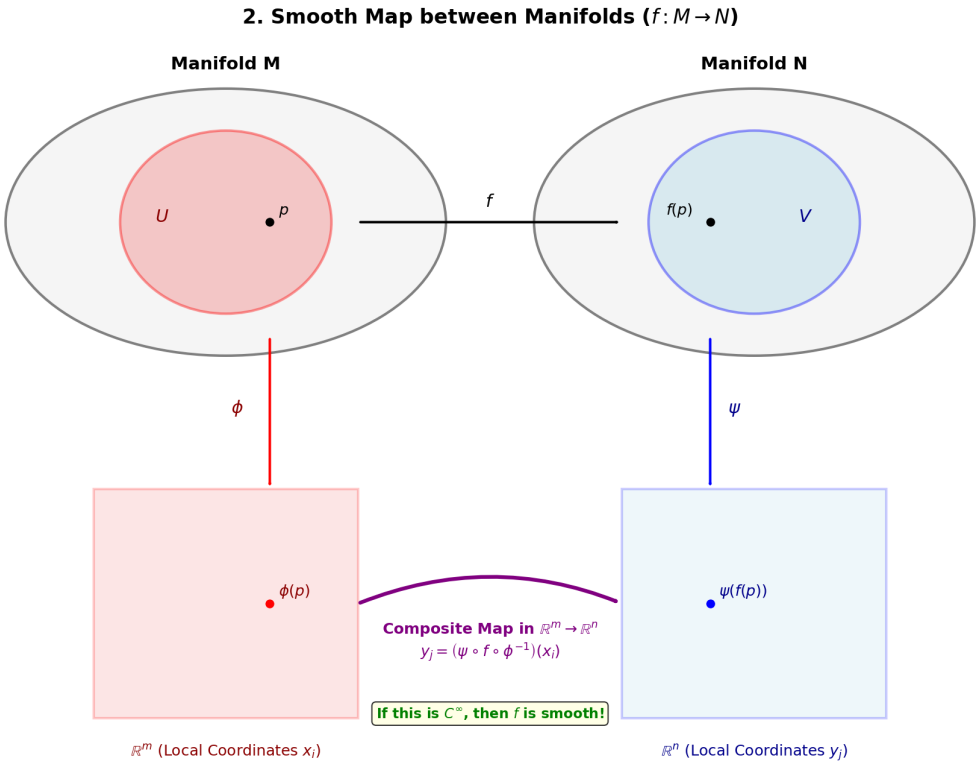
- **For $n = 4$:** No! There exist infinitely many **Exotic $\mathbb{R}^4$s**. These are manifolds topologically identical to standard 4D Euclidean space but *not* smoothly equivalent to it. This anomaly is of immense interest since our physical universe is modeled as a 4-dimensional spacetime.

1.7 Smooth Maps and Diffeomorphisms

Consider a mapping $f : M \to N$. To determine if f is smooth, we project it down into local flat coordinates where we can do calculus. We construct a composite function strictly in real Euclidean space:

$$\psi \circ f \circ \phi^{-1} : \mathbb{R}^m \to \mathbb{R}^n$$

We define the map f to be a **smooth map** ($f \in C^\infty$) if and only if this multivariable real function is infinitely differentiable.



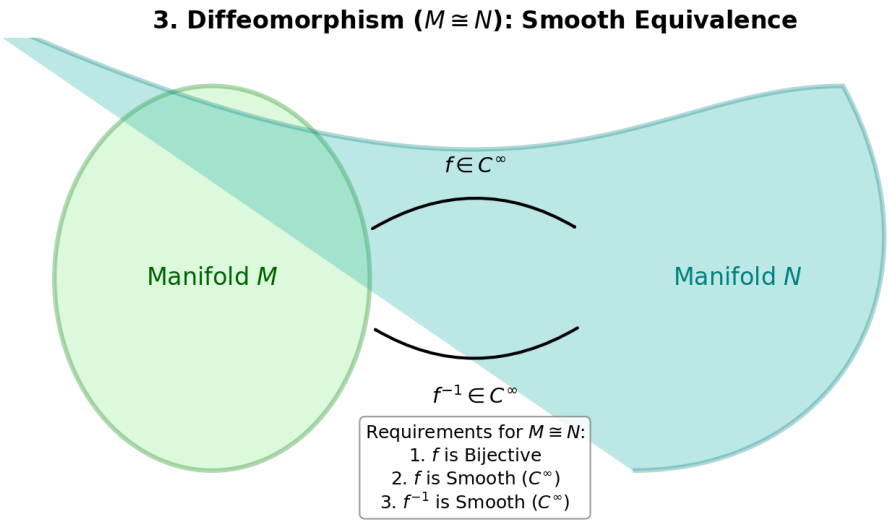
(a) Smooth Map $f : M \to N$

1.7.1 Diffeomorphism: The Ultimate Equivalence

Two differentiable manifolds M and N are diffeomorphic ($M \cong N$) if there exists a mapping $f : M \to N$ satisfying three conditions:

1. **Bijection:** The map f is both injective and surjective.
2. **Smooth Mapping:** $f \in C^\infty$.
3. **Smooth Inverse:** The inverse map $f^{-1} \in C^\infty$.

If two spaces are diffeomorphic, they are indistinguishable from the perspective of differential geometry.



(b) Diffeomorphism $M \cong N$

1.8 Calculus on Manifolds: Defining the Derivative

How do we take a derivative on an abstract curved manifold with no inherent flat grid? Consider a scalar field: $f : M \to \mathbb{R}$. We want the partial derivative of f at a point $p \in M$.

We "pull back" the computation to flat space using a local coordinate chart ϕ:

1. Map the abstract point p to a coordinate vector in $\mathbb{R}^n$: $x = \phi(p)$.
2. Construct the composite function mapping $\mathbb{R}^n \to \mathbb{R}$: $f \circ \phi^{-1}$.
3. Take the standard partial derivative of this composite function in flat space.

Mathematically, the derivative with respect to the coordinate x^μ at point p is defined as:

$$\left.\frac{\partial f}{\partial x^\mu}\right|_p = \frac{\partial (f \circ \phi^{-1})}{\partial x^\mu}(\phi(p))$$

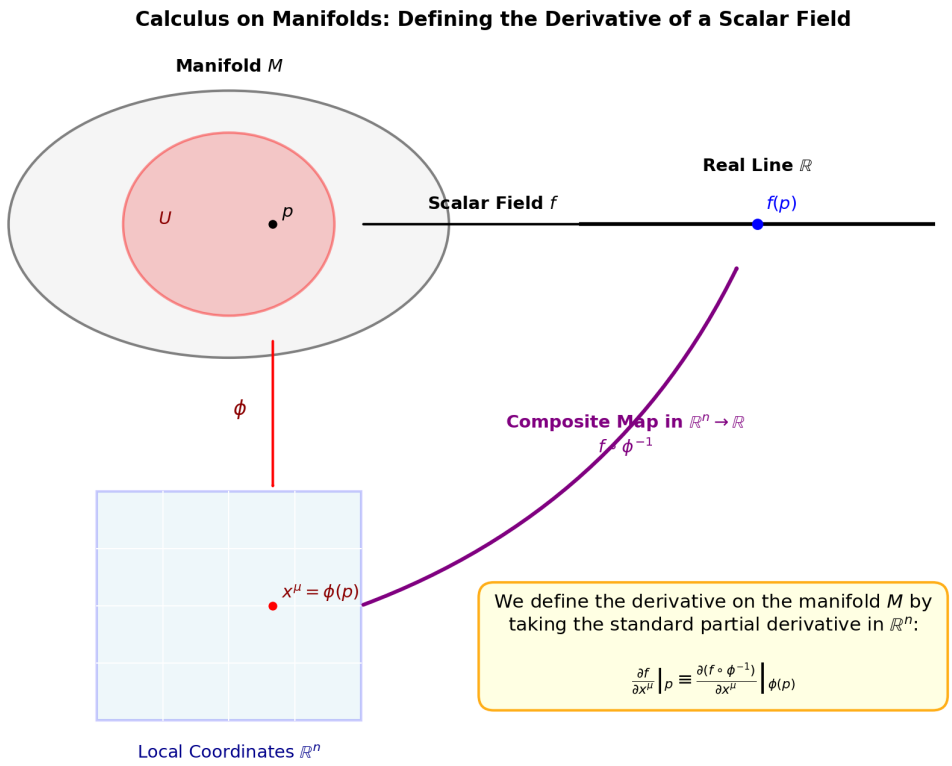


Figure 1.3: Calculus on Manifolds: Defining the derivative via local coordinates.

textbfA Critical Observation: This definition successfully allows us to compute rates of change, but it explicitly depends on the choice of the coordinate chart ϕ. In the next lecture, we will resolve this coordinate-dependence by introducing tangent vectors and the tangent space.